

German Electricity Demand Forecasting Case Study

Advanced Data Science - Time Series Modelling

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1. Objective and modelling design:

In this work we apply several forecasting methods to forecast the electricity demand in Germany, ranging from simple benchmarks to statistical, feature-based and neural network models. The objective is to determine whether the models are usable in practice apart from minimizing the error. Electricity demand projections are important because grid operators need to balance generation and consumption, but also to avoid unnecessary spending on reserves and reliability problems.

Data were collected from the hourly time-series file of the Open Power System Data. German real load column, DE_load_actual_entsoe_transparency, was split and filtered from 01.01.2015 until the end of available 2020 records. For the exploratory analysis and benchmark, SARIMA, SARIMAX and Random Forest models, we generated the daily and weekly average series. For the LSTM experiment, we directly used the hourly data. The last 104 weekly observations were held out for a two-year test period.

2. Data preparation and preliminary results:

The load column was renamed to a more descriptive variable name, and the timestamp was transformed into a time-series index. The cleaned working data set included 50,400 hourly, 2,100 daily, and 301 weekly observations. No further missed German load observations after preparation. Weekly averaging was selected for most models, as it preserves the annual seasonal trend, and reduces short-term hourly noise.

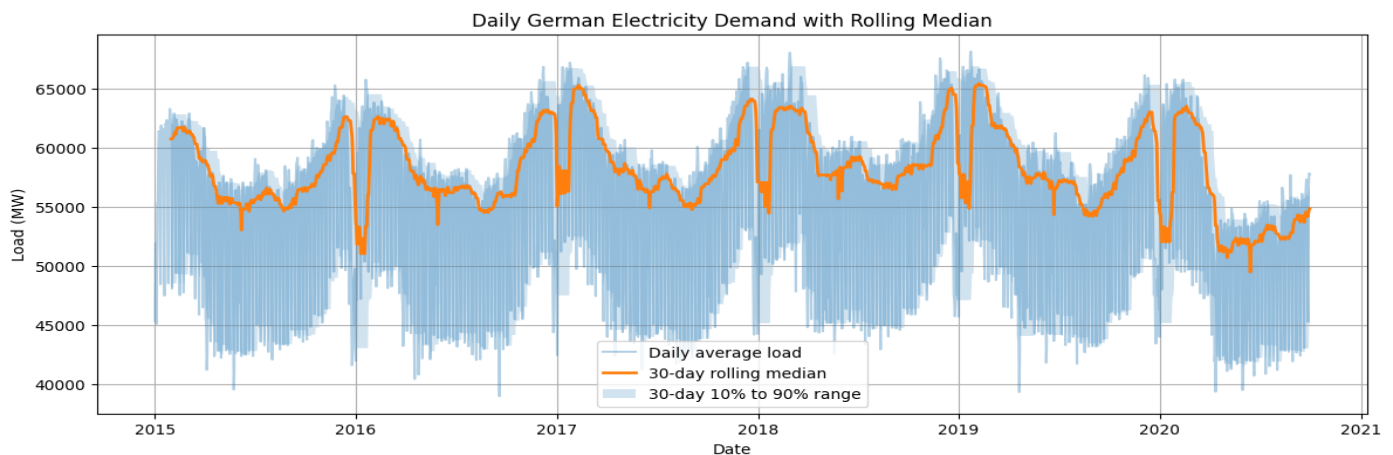


Figure 1. Daily electricity demand with rolling median and variation band.

The exploratory analysis showed that there were annual patterns of activity rather than random movement. Generally, demand was higher in the colder months and lower in warmer ones. The load duration curve shows a high base level of demand for most of the time and fewer high-load times that are important for peak planning. The monthly seasonal indicator suggests that demand levels fluctuate throughout the year.

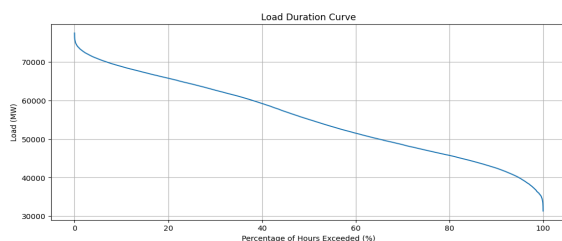


Figure 2a. Load duration curve.

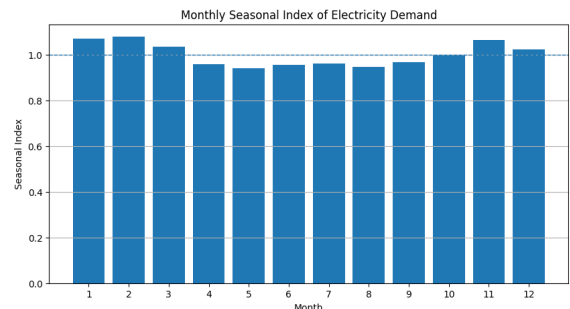


Figure 2b. Monthly seasonal index.

3. Autocorrelation and stationarity:

The ADF and KPSS tests address stationarity from different perspectives, so they were used together. The weekly load was statistically suitable for modelling with a p-value of ADF 0.00118 and a p-value of KPSS 0.100. Moreover, a first difference was tested and provided more convincing evidence of stationarity. However, the seasonal structure was maintained as an important feature of the modelling as the ACF/PACF plots still showed seasonal dependence.

Series	ADF statistic	ADF p-value	KPSS statistic	KPSS p-value	Interpretation
Weekly load	-4.047	0.00118	0.160	0.100	Stationary by both tests
First difference	-7.069	<0.001	0.056	0.100	Stronger stationarity

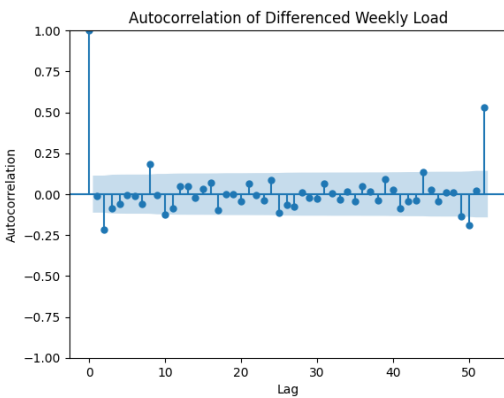


Figure 3a. ACF of differenced weekly load.

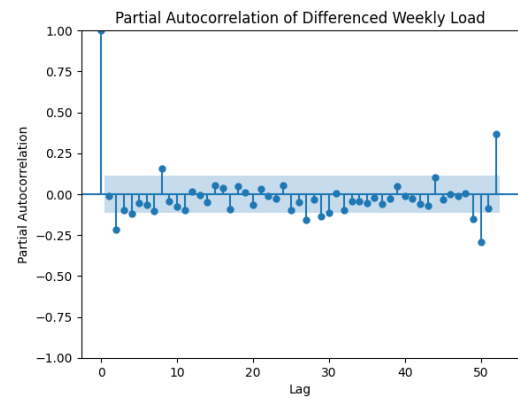


Figure 3b. PACF of differenced weekly load.

4. Forecasts for benchmark's Models:

Prior to the advanced models, four benchmark forecasts, mean, naive, seasonal naive, and drift—were fitted. These are crucial since a sophisticated model can only be valuable if it outperforms a straightforward baseline. The best benchmark was the seasonal naïve approach, which replicates the same week from the prior year.

Benchmark model	RMSE	MAE	MAPE (%)
Seasonal naive	3006.76	2318.52	4.41
Mean forecast	4397.30	3788.83	6.97
Naive forecast	4459.11	3783.20	6.79
Drift forecast	5117.96	4339.89	8.05

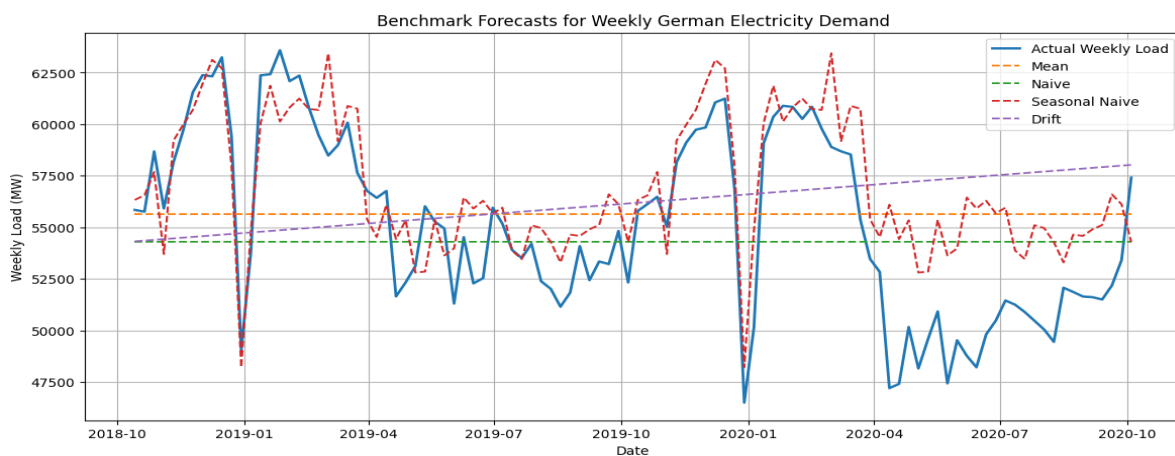


Figure 4. Benchmark forecasts against observed weekly demand in the two-year test period.

The benchmark data shows a very strong yearly recurrence of German power demand. Linear trend was not appropriate because the drift model had the worst performance. Subsequent models used the seasonal naïve outcome as their main benchmark.

5. The SARIMA Model:

The weekly data showed autocorrelation and seasonality over the year, so a SARIMA model was suitable. The seasonal period was chosen to be 52 weeks. The necessary grid search across $p = 0$ to 6, $d = 0$ to 2, and $q = 0$ to 6 found 147 non-seasonal parameter combinations. The final forecast Version 3 was SARIMA (4,2,6) (1,1,1,52). However, the best likelihood choice by AIC is SARIMA (2,2,6) (1,1,1,52).

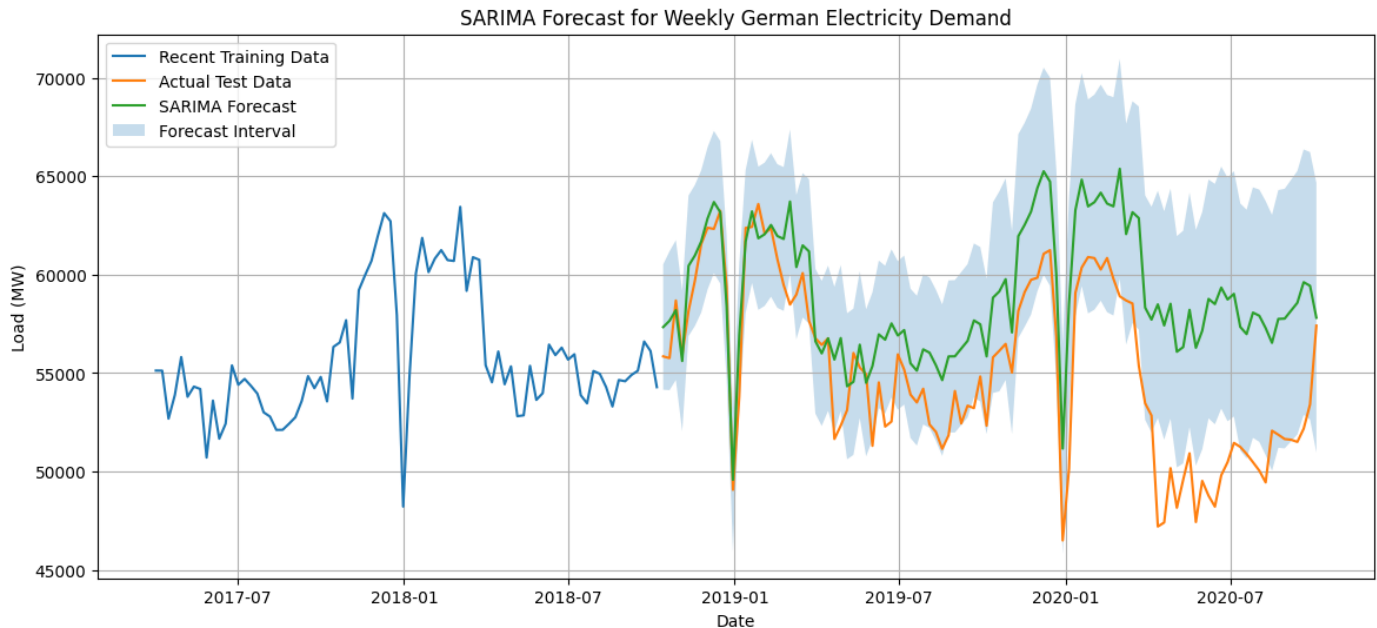


Figure 5. Final SARIMA forecast with confidence interval.

Model	Frequency	RMSE	MAE	MAPE (%)
Seasonal naïve	Weekly	3006.76	2318.52	4.41
SARIMA	Weekly	4765.23	3883.62	7.37

The seasonal naïve benchmark outperformed the SARIMA model. This conclusion is helpful because it demonstrates that a strong hold-out forecast is not always the consequence of a solid likelihood fit. There may have been less generalization during the test time due to over-differencing and atypical behavior. To determine whether structure persisted after modelling, the residual histogram and residual ACF were examined.

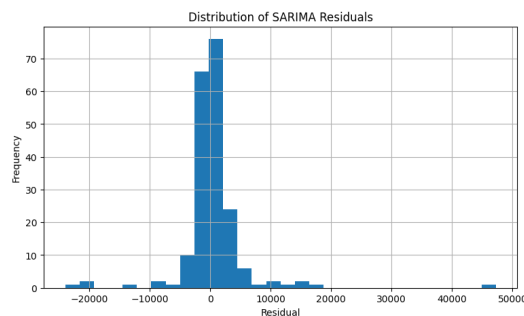


Figure 6a. SARIMA residual distribution.

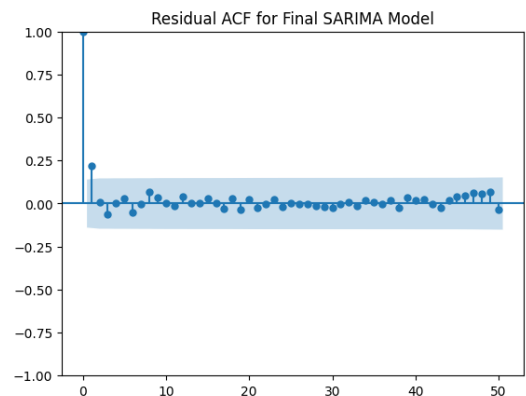


Figure 6b. SARIMA residual ACF.

6. SARIMAX with holiday and weather variables

In a SARIMAX model, the temperature in Berlin and the number of German holidays were included as exogenous variables. Temperature was added since holidays can alter commercial and industrial activity, and heating and cooling behavior might impact load. Although future observed temperatures are unknown at the forecast origin unless they are derived from a weather forecast, holiday variables are known beforehand. As a result, the SARIMAX result should be referred to as an explanatory or conditional forecast.

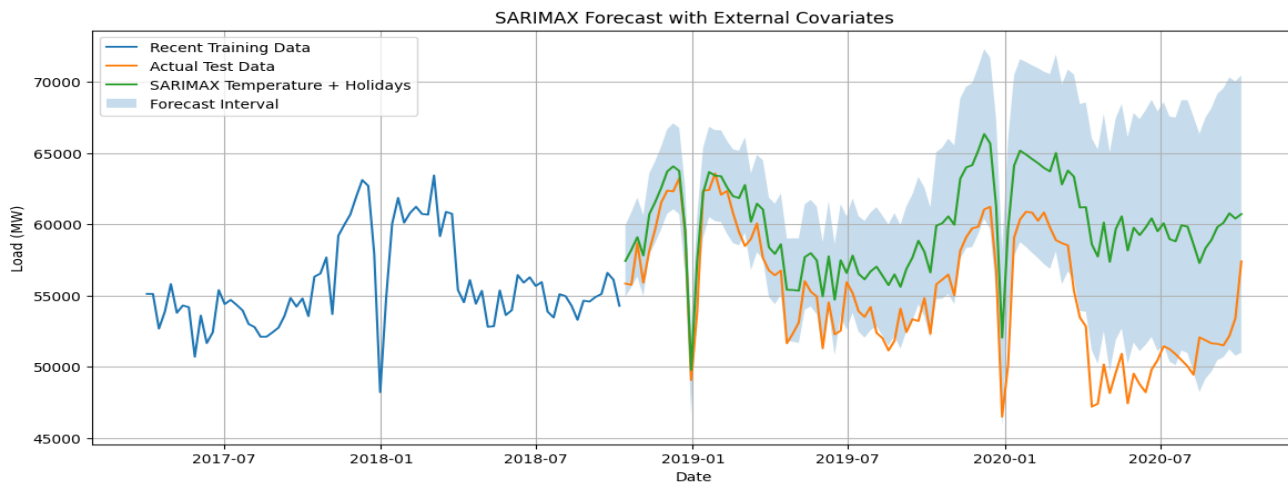


Figure 7. SARIMAX forecast using Berlin temperature and holiday covariates.

Model	Frequency	RMSE	MAE	MAPE (%)
Seasonal naive	Weekly	3006.76	2318.52	4.41
SARIMAX temperature + holidays	Weekly	5633.88	4712.47	8.92

The additional covariates in this version did not increase accuracy. This implies that the SARIMAX structure did not make good use of temperature and holidays, but it does not imply that they are worthless. Additionally, a single Berlin temperature series is merely a stand-in for the national weather of Germany. It would be preferable to use a population-weighted temperature index from multiple cities.

7. The Random Forest feature model:

A Random Forest regressor with lagged load values, rolling statistics, calendar variables, weather characteristics, and holiday indicators was employed in the feature-based model. Only data that was accessible prior to the projected week was utilised by shifting the lag and rolling characteristics. This was crucial to preventing data leaks. Random Forest does not need the weekly series to be stationary in order to model non-linear relationships.

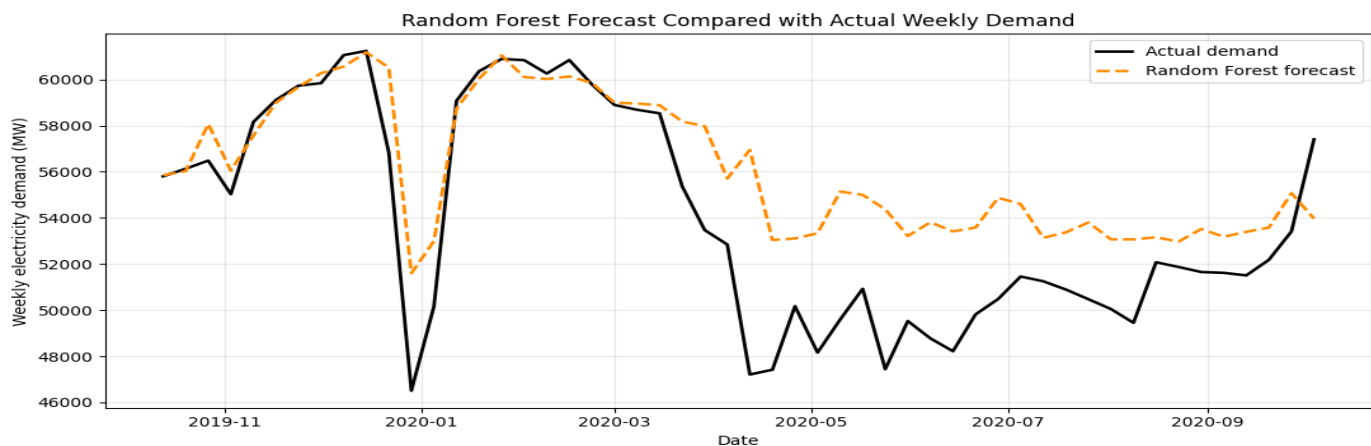


Figure 8. Random Forest weekly forecast against actual demand.

The best weekly model was Random Forest. For the seasonal naïve benchmark, it lowered MAPE from 4.41% to 3.51%. Lagged demand, particularly the 52-week lag, dominated feature relevance, indicating that load from the prior year is a significant predictor of future demand.

Model	Frequency	RMSE	MAE	MAPE (%)
Seasonal naïve	Weekly	3006.76	2318.52	4.41
Random Forest	Weekly	2505.60	1826.49	3.51

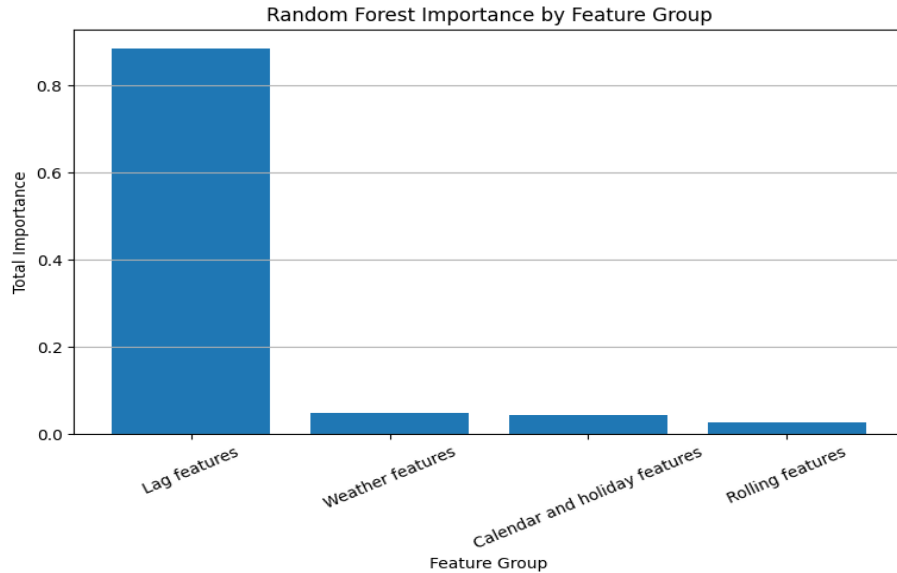


Figure 9. Grouped feature importance from the Random Forest model.

8. The hourly LSTM model:

The initial hourly demand data was used to train the LSTM model. As a result, the model was able to learn weekly and daily temporal patterns that are hidden after weekly aggregate. To prevent leakage from subsequent observations, the scaler was only installed during the training phase and then applied during the test period. Before choosing the final model, a quick tuning check contrasted different lookback and layer settings.

LSTM setting	Hours	Layer design	Dropout	Epochs	RMSE	MAPE (%)
Weekly Memory LSTM	168	50 + 50	0.20	8	1690.02	2.40
Two Week Memory LSTM	336	64 + 32	0.25	4	2247.75	3.17
Short Memory LSTM	48	32	0.20	4	2231.35	3.09

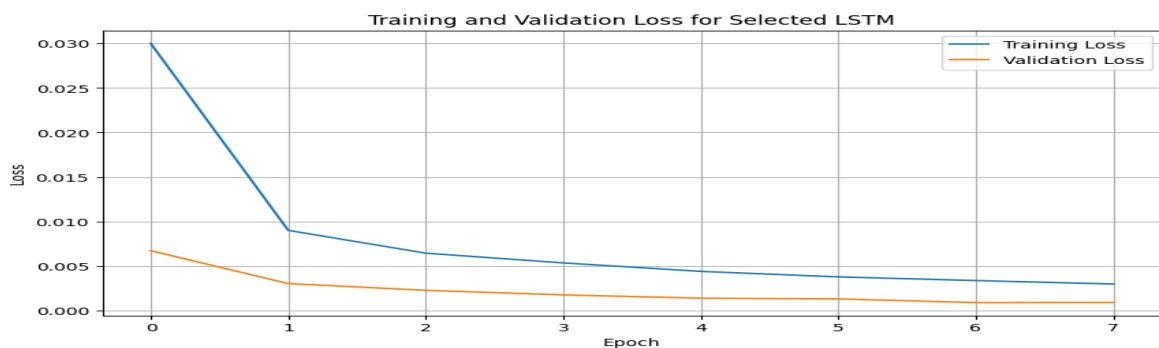


Figure 10. Training and validation loss for the selected LSTM model.

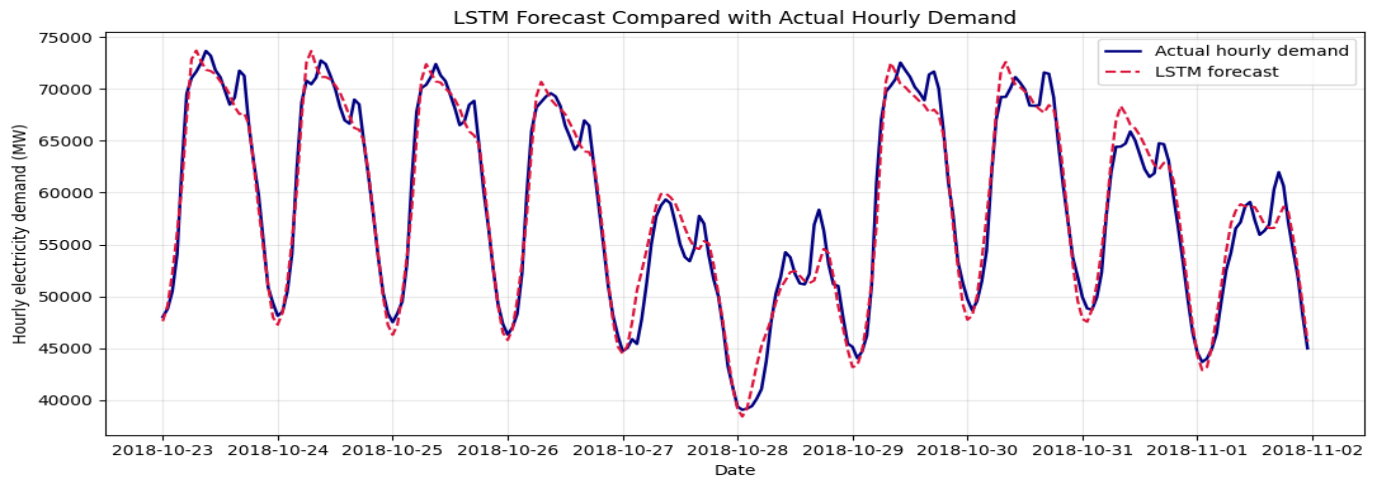


Figure 11. LSTM forecast sample against actual hourly demand

With a MAPE of 2.40%, the LSTM produced the lowest percentage error. SARIMA, SARIMAX, and Random Forest are weekly models, whereas this outcome is assessed using hourly observations. Therefore, if hourly demand forecasting is the operational goal, the LSTM is most pertinent. Compared to the Random Forest model, it is also harder to understand and takes more upkeep.

9. Answers for the Questions about Analysis:

9.1. Compare all models against the seasonal naive benchmark. Which models, if any, provide a meaningful improvement?

Answer:

German electricity demand has a clear yearly pattern and the seasonal naive model served as a strong benchmark. It had RMSE 3006.76 and MAPE of 4.41%. For weekly data Random Forest was the best model reducing RMSE to 2505.60 and MAPE to 3.51%. Therefore, lagged demand, calendar variables and engineered features provide additional information over just repeating the value from the same week in the previous year. In this SARIMA and SARIMAX did not outperform the seasonal naive benchmark. They find that a model may fit the training pattern but may not generalise well over the last two-year period of testing. The LSTM resulted in a lower MAPE (2.40%) but used hourly data and should therefore not be compared directly to the weekly models without acknowledging the different frequency and forecasting set-up.

9.2. Explain how you avoided data leakage when creating temperature lag features?

Answer:

We avoided data leakage by making sure that lagged and rolling features only used past information. Load lags were shifted before modelling, but the target week was not included as an input. The rolling variables were also calculated from past observations instead of the current or future target value. Lagged temperature features were used for temperature where possible, so the model did not rely solely on weather in the future. This is important because the actual future temperature would not be known at the time of the original forecast. When I used observed test-period temperature in SARIMAX, I considered that result to be a conditional/explanatory forecast. Thus, the SARIMAX weather result assumes future temperature is known from observed data or a weather forecast, so it is not a fully operational forecast by itself.

9.3. For the SARIMAX model, justify the chosen differencing orders and seasonal period?

Answer:

The seasonal period was set to 52 weeks, because the weekly demand data had a repetitive structure every year. This choice is consistent with the data frequency and the annual electricity demand cycle as identified in the exploratory plots and autocorrelation patterns. The differencing choices were based on stationarity tests, ACF/PACF plots and SARIMA search based on AIC. ADF and KPSS results indicated that the weekly series was weakly non-stationary but the seasonal pattern was still obvious. Hence the seasonal differencing was done to help the model to deal with the annual persistence. But the last forecast performance was worse than the seasonal naive model. This suggests the differencing and the SARIMA structure are too complex or too aggressive for this test period.

9.4. Evaluate whether temperature and holiday covariates improve forecast accuracy. Are these covariates known at the forecast origin?

Answer:

Adding temperature and holiday covariates did not improve SARIMAX over the seasonal naive benchmark. The RMSE and MAPE of the SARIMAX model with temperature and holidays was 5633.88 and 8.92%, respectively, worse than the seasonal naive result. So, the outside covariates were not enough to overcome the weakness of the SARIMAX structure in this run. The dates of public holidays are known in advance; thus holiday variables are safe for operational forecasting. temperature though. For the forecast origin the actual future temperature is not known, unless a weather forecast is available. Therefore, the SARIMAX results, which are based on temperature, should be interpreted as conditional forecasts, rather than as fully operational real-time forecasts.

9.5. Compare the interpretability and complexity of SARIMAX, feature-based and neural network models?

Answer:

SARIMAX is the most interpretable model as the autoregressive, moving-average, differencing, seasonal and exogenous parts can be explained directly. It also gives confidence intervals which are useful for uncertainty. But this version was not strong on accuracy. Random Forest is less transparent than SARIMAX but it still provides feature importance so it is possible to understand which variables are useful. It can also model non-linear dependencies and does not need the same stationarity assumptions of SARIMA/SARIMAX. The least interpretable and most complex model is the LSTM. It can learn detailed hourly patterns, but needs more preprocessing, tuning and computing time. It is also more difficult to explain to a non-technical reader. Therefore, LSTM is useful for high-frequency demand forecasting, but it is more difficult to maintain.

9.6. Recommend one model for operational use and justify your choice using accuracy, uncertainty, interpretability and maintenance considerations?

Answer:

For weekly operational forecasting I would propose the Random Forest model in this version. The seasonal naive benchmark was bettered from 3006.76 to 2505.60 in RMSE and from 4.41% to 3.51% in MAPE. It also has lagged demand and calendar features, which come in handy for forecasting. Random Forest is less interpretable than SARIMAX, but the feature importance can still provide a useful explanation of the model. It is also easier to maintain than LSTM as it requires less tuning and easier to re-train. SARIMA and SARIMAX are helpful to interpret and uncertainty intervals, but they were not accurate enough here. The LSTM model worked well in hourly forecasting but is too complex for the weekly operational recommendation. Hence, Random Forest is the best compromise while seasonal naive remains a simple backup benchmark.

10. Final comparison and interpretation:

Model	Frequency	RMSE	MAE	MAPE (%)
LSTM	Hourly	1690.02	1282.10	2.40
Random Forest	Weekly	2505.60	1826.49	3.51
Seasonal naive	Weekly	3006.76	2318.52	4.41
SARIMA	Weekly	4765.23	3883.62	7.37
SARIMAX temperature + holidays	Weekly	5633.88	4712.47	8.92

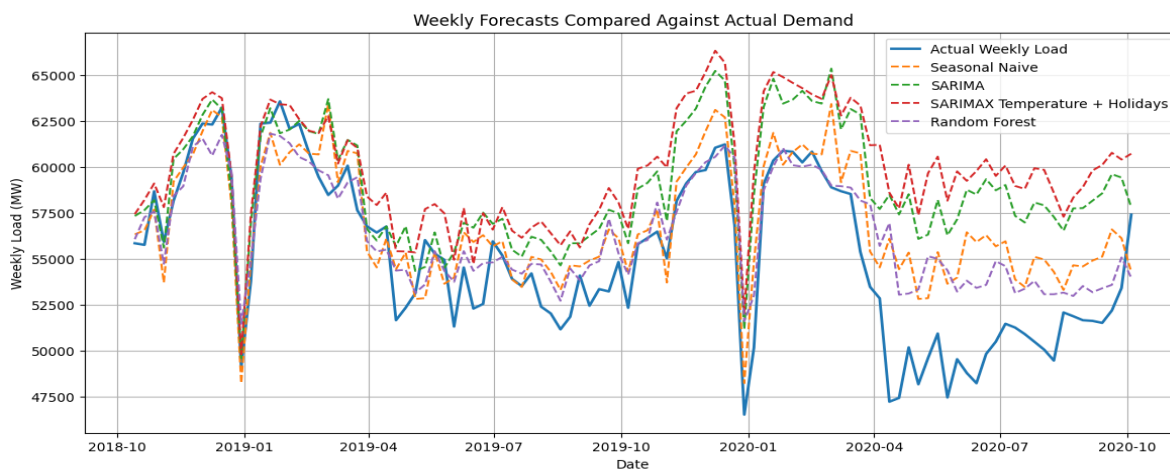


Figure 12. Final weekly forecast comparison.

Random Forest showed a significant weekly improvement over the seasonal naive benchmark. In this run SARIMA and SARIMAX could not beat the seasonal naive benchmark. The LSTM used hourly data and had the lowest overall error, but it cannot be directly compared with the weekly models until its predictions are aggregated to weekly values.

The rolling and temperature lag characteristics were purely historical, which minimized the risk of leakage. The SARIMAX temperature projection is still conditional, as observed future temperatures were used in the test period. Holidays are safer operational covariates as the dates of the holidays are known before the forecast. Weekly data and annual repetition supported the selected SARIMA seasonal period of 52, stationarity checks and the AIC search led to differencing.

In this case SARIMAX has low accuracy but is the most interpretable and gives confidence ranges. Random Forest performed better than the other weekly models but is less transparent while still prioritizing features. LSTM is the most complicated and difficult to understand but can learn high frequency patterns from hourly data. Random Forest is the best balanced option for operational weekly forecasting, as it uses mainly known or lagged features, boosts accuracy and is easier to retrain than a neural network. Seasonal naive should remain the simple backup standard.

11. Limitations, conclusion, and future work:

The study has shown that there is a strong yearly and calendar-related pattern in German electricity demand. The seasonal naive standard was hard to beat, indicating the importance of demand in previous years. The best hourly result was from the LSTM model while the best weekly model was the Random Forest model. For this Version 3 experiment SARIMA and SARIMAX did not generalise as well as we would have hoped, though they were useful for interpretation and uncertainty.

The main limitations are the fixed train-test split, the use of the temperature of Berlin as a proxy of the national one, and the difficulty of predicting the 2019-2020 period from the past data. Future research should include multiple German meteorological stations, rolling-origin cross-validation, probabilistic machine learning forecasts and other neural architectures such as CNN-LSTM or GRU.

12. References:

1. Analysis of Time Series: Forecasting and Control Box, G.E.P., Jenkins, G.M., Reinsel, G.C. and Ljung, G.M. (2015). Hoboken: Wiley, 5th ed. P. 9781118675021 TIME SERIES EVALUATION FORECASTING AND CONTROL 5TH EDITION, GENERAL INTRODUCTORY STATISTICS, Wiley
<https://www.wiley.com/en-us/shop/general-introductory-statistics/time-series-analysis-forecasting-and-control-5th-edition-p-9781118675021>
2. Dickey, D. A. "Distribution of Estimators for Autoregressive Time Series with a Unit Root. and Fuller, W. A. Journal of the American Statistical Association 74(366a): 427-431. (1979).
<https://www.tandfonline.com/doi/abs/10.1080/01621459.1979.10482531>
3. "Testing the null hypothesis of stationarity against the alternative of a unit root." Kwiatkowski, D., Phillips, P.C.B., Schmidt, P. and Shin, Y. (1992). Journal of Econometrics, 54, 159–178. Use this for the KPSS test. This is the original document for KPSS Stationarity Test.
https://econpapers.repec.org/article/eeeeconom/v_3a54_3ay_3a1992_3ai_3a1-3_3ap_3a159-178.htm
4. Breiman, L. Random forests. (2001) Machine Learning, 45, pp. 5-32. This is for your feature based Random Forest model. This is the original Random Forest paper, a trusted academic source.
<https://link.springer.com/article/10.1023/A:1010933404324>
5. Hong, T. "Probabilistic electric load forecasting: A tutorial review," International Journal of Forecasting 32, 3 (2016): 914-938. and S. Fan. (2016). Use it to explore operational forecasting, uncertainty and the context of electrical load forecasting. This evaluation of load forecasts shall be subject to peer review.
<https://research.monash.edu/en/publications/probabilistic-electric-load-forecasting-a-tutorial-review/>
6. Fan, S. and Hyndman, R. J. (2012). 'Short-term load forecasting based on a semi-parametric additive model', IEEE Transactions on Power Systems, 27(1), pp. 134–141.
<https://doi.org/10.1109/TPWRS.2011.2162082>